Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/03/2006
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Project Title:	Stress-Driven Instabilities on Helium-4 Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	02/01/2001	**End Date:**	10/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: is on NCE to 10/30/2006, per N. Hall-GRC (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2868		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover the mysteries of the universe obscured by Earth's gravity		

Task Description:	Helium-4 is pressurized at low temperatures near 1K to grow a crystal in a rectangularly shaped chamber. After a solid seed is formed, it is slowly grown to fill the lower part of the chamber. When the crystal is grown to the height of about 5 mm, the temperature and the pressure are stabilized. The crystal is then put under an externally applied non-hydrostatic stress. The external stress is applied by the use of piezoeletric piston. The interface between the liquid melt and crystal is observed as a function of the applied stress, time and temperature. Optical interferometry techniques are employed to observe the surface profile. If the applied stress is sufficiently small, the solid height is expected simply to decrease in proportion to the square of the applied stress. Interesting instabilities are expected to occur at the interface when the applied stress is large enough to exceed a certain threshold level. The instabilities under a variety of conditions of the stress direction, crystal thickness, boundary conditions and temperature are being studied. Our objective of this project is to study the surface profile of helium-4 crystals when they are placed under externally applied stress. We wish to observe the expected corrugations of surface shape and the dynamics in the surface profile developement under different applied stress conditions and to compare these observations with predictions based on the theories of stress-driven instabilities. The formation and growth of crystal shapes are fundamental problems in general. Helium crystals at low temperatures are chosed as the test material to study these important problems. Quantum crystals of helium-4 provide an excellent testing ground for the stress-driven instability effects owing to their high purity, temperature homogeneity, and rapid crystal growth over macroscopic sizes. The superfluid liquid melt provides the mechanisms for rapid heat dissipation, freezing and melting which allow observations of the dynamic processes over short time scales. We believe our studies of stress-driven instability can contribute significantly to better understanding of the problems. Understanding of stress-driven instability is also valuable for its close relation to the phenomena of practical importance such as pattern formation in epitaxial crystal growth, surface reaction rate on pre-stressed solids and surface cracking and corrosion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During 2004 grant period, a cryogenic optical interferometric apparatus for observing the surface profile of helium-4 crystal was completed and preliminary tests were carried out. Briefly, a He-Ne laser beam is guided into the vacuum can to illuminate the 17 mm by 10 mm cross section of the helium crystal growth chamber. A portion of the beam is partially reflected from the front window of the chamber. The rest of the beam passes through the crystal chamber, passes onto the "sample" mirror and reflected back. The two beams reflected from the front window and sample mirrors are superposed to produce interference. The interference image pattern is transmitted to a room temperature recording system with a CCD camera. To form solid in the chamber, the liquid pressure is raised to about 24.5 bars and the temperature is stabilized at 1.20 K. The pressure is then slowly increased until a solid seed forms. The first time a solid is grown, a pressure greater than the equilibrium melting pressure (25 bars) is usually needed before a seed appears. The presence of a solid meniscus around the seed is can be seen in the image. The seed can be easily manipulated to increase or decrease in size. The movable piston is located at the left side of the figure. The pattern outside of the seed is determined by the background interference. The seed is shrunk to a small size and “annealed” for an hour and then grown back slowly over two to three hours to a vertical thickness of 3 ~ 4 mm over the entire bottom surface. Subsequently, stress is exerted onto the solid by moving a piston attached to a PZT to which an external voltage Vpzt is applied. While Vpzt is less than a threshold value (1800 ~ 2300 V), the interference pattern shows locally uneven but smooth variation of height. In general, the solid height decreases in the region near the piston. The height changes by lesser amounts or even increases in the region far away from the piston depending on the particular solid sample. According to the linear equilibrium response between liquid and solid under strain, the solid height is expected to fall globally. Our observations indicate that the applied strain on the solid is not uniformly transmitted across the entire length of the sample. The boundary condition between the solid helium and the chamber wall (machined copper at present) requires further studies. If Vpzt does not exceed the threshold value for the sample solid, the interference pattern reverts reversibly to the original pattern when Vpzt is decreased back to zero. When Vpzt exceeds the threshold value for a given sample, spectacular localized patterns develop. We thus observe an onset of instability expected on theoretical grounds. When Vpzt is increased to 4000 V, the stress induce several regions to form deep valleys whose nascent stages can be seen by the interference line pattern curling up into high line density regions. The interference pattern relaxes while Vpzt is held constant. This interesting relaxation effects will be investigated further in the future. The interference patterns are no longer reversible as Vpzt is decreased, and different pattern develops if same time sequence of Vpzt is repeated. The threshold and interference pattern are different for each solid samples. The orientation of crystal axis of our samples relative to relative to PZT is not controlled. This might explain the oblique angles of the deformations produced.
Bibliography Type:	Description: (Last Updated: 01/05/2007)
Abstracts for Journals and Proceedings	Kojima H, Grinfeld P. "Observation and Simulation of Stress Driven Instabilities on Solid Helium Surface." 24-th Int. Conf. Low Temperature Physics, Orlando, FL, August 10-17, 2005. Proceedings of 24-th Int. Conf. Low Temp. Phys., August 2005. , Aug-2005